

The Crypto Lending Platform **That Explains Itself**

Every other platform is building faster pipes. Aetherum gives your members a lending decision they can actually understand — and that explanation is the moat.

[Schedule a Demo](#)[Download PDF](#)

Built for the Member. Deployed for the Credit Union.

The market has been focused on speed for the institution. Aetherum is focused on trust for the member.



DACS-LTV

PATENT PENDING

Crypto-Backed Lending

Your members keep their crypto exposure, avoid a taxable event, and get USD liquidity — all through their trusted credit union. Aetherum handles the infrastructure. You own the relationship.



PATENT PENDING

DACS Risk Assessment

Our patent-pending ten-pillar risk model evaluates crypto collateral in real time. Every decision comes with a plain-language explanation the member can read and understand — not just a number on a screen.



TAX INTELLIGENCE

PATENT PENDING

Tax-Position Score & Collateral Optimizer

Aetherum's TPSOE engine scores the tax efficiency of each collateral position and ranks pledge assets to minimize member tax liability. The Collateral Optimizer selects the optimal set for any loan request — a feature no D2C competitor offers institutions.



Smart Review



RWA Tokenization



PATENT PENDING

AI-Powered Underwriting Guidance

Before a member signs, Aetherum evaluates their full borrowing position and explains it in plain language. Recommended amount, risk level, and reasoning — delivered instantly, automatically, at the point of decision.

Tokenize real-world assets like real estate, receivables, and treasury instruments to expand collateral options and create new secondary-market liquidity for credit unions.

XLS-66 LoanBroker (XRPL Native Lending)

Aetherum operates as the LoanBroker in Ripple's XLS-66 Lending Protocol — managing DACS-gated loan origination, dual-party transaction authorization (BIDSF), and First-Loss Capital as a regulatory capital analog for NCUA-examined institutions. RLUSD as native loan currency. Live on XRPL testnet.



LIVE INTEGRATION

CIRCLE CCTP

Pharos Partnership Panel

Live Pharos Atlantic Testnet integration — USDC contract wired, wallet generation, balance checker, and USDC disbursement demo. Powered by Circle CCTP.



OPT-IN INTELLIGENCE

PATENT PENDING

Member Crypto Insights

Real-time visibility into the digital assets your members already own — before they need a loan. Consent-based wallet connections, DACS eligibility scoring, and opt-in loan pipeline. See what's invisible today.



ERC-3643

PATENT PENDING

On-Chain Member Identity

ERC-3643 compliant identity verification for every borrower. Each member receives a portable ONCHAINID attestation — KYC verified, jurisdiction gated, and collateral-scored on-chain. Built on the T-REX standard.



PRIVACY-PRESERVING

PATENT PENDING

Privacy-Preserving Underwriting System (FHE-DACS)

Encrypted risk assessment computation using homomorphic encryption — member financial data is never exposed in plaintext during underwriting. Patent Pending.



DACS-LTV

PATENT PENDING

Dynamic LTV Enforcement

DACS risk tier automatically sets each member's on-chain loan-to-value ceiling — enforced at the blockchain layer via ERC-3643. Tier 1 members access up to 70% LTV, Tier 2 up to 65%, Tier 3 up to 55%.



PATENT PENDING

Distributed Ledger Collateral Management & Liquidation Control

A distributed ledger architecture that tracks collateral positions, enforces margin calls, and executes liquidation logic on-chain — giving credit unions a tamper-proof, auditable record of every collateral event from origination to resolution.



ERC-3643

On-Chain Identity Attestation

Automated on-chain attestation triggered by DACS scoring — members are registered as verified credit union participants directly on Sepolia and Pharos Atlantic Testnet via ERC-3643 compliance modules. US jurisdiction verification and collateral LTV are set on-chain in real time.



PATENT PENDING

Collateral Intelligence Engine (CIE™)

Every collateral position monitored 24/7. Finality-aware on-chain event processing, live LTV computation, DACS score updated on every price movement, and institution-authorized liquidation control — all automated. The always-on layer that makes crypto collateral operationally viable for credit union loan officers.



LIVE

PATENT PENDING

Native LTV Monitoring

Real-time BTC and ETH collateral wallet surveillance polling every 60 seconds via Blockstream and Alchemy — no third-party monitoring dependency. 30-day rolling volatility feeds directly into DACS. Filed May 26, 2026 — Application #64/075,187.



LIVE

PATENT PENDING

Liquidation Escalation Layer

Three-tier human oversight: Tier 1 auto-executes standard positions, Tier 2 pages your risk officer with a 30-minute override window, Tier 3 requires explicit loan officer authorization for \$250K+ or 40%+ price drops. Every action timestamped for NCUA audit. Filed May 26, 2026 — Confirmation #6958.



SECURITY REVIEW

DDQ Engine — 35-Entry Knowledge Base

Built-in due diligence questionnaire engine with 35 pre-approved KB entries covering the full SIG security review framework — custody, encryption, incident response, key management, third-party risk, and more. IT and risk teams get instant, compliance-vetted answers without waiting for a sales call.



Staking Module

Members earn yield on supported digital assets directly through the platform while maintaining collateral eligibility for crypto-backed loans.



COMING SOON

Payments / Wallet Transfers

Seamless on-chain and off-chain payment flows that let members



Trading

Integrated digital asset trading that lets members buy, sell, and rebalance their crypto portfolio directly through the Aetherum



PATENT PENDING

Collateral Intelligence Engine

Real-time collateral surveillance for every active loan. Finality-

move digital assets between wallets, fund loan repayments, and initiate transfers — all within your credit union's platform.

platform without leaving your credit union's ecosystem.

aware on-chain event processing, live LTV computation, DACS score updates on every price movement, and institution-authorized liquidation control — all automated.



Tax Savings

Automated tax-loss harvesting, gain/loss reporting, and collateral optimization tools that help members minimize their crypto tax liability at every step of the loan lifecycle.



Fraud Detection

AI-powered real-time fraud scoring and behavioral analytics monitor every transaction and onboarding event to protect your credit union and your members from bad actors.



Claims Process

Structured workflows for collateral disputes, liquidation events, and member escalations — giving your team a clear, auditable path from incident to resolution.



Loan Officer Portal

A dedicated dashboard for loan officers to review applications, assess DACS scores, manage active loans, and action margin call alerts — purpose-built for crypto-collateralized lending.



Compliance Officer

Siloed KYC/AML screening queue, sanctions monitoring, and audit-ready reporting tools that give your compliance team full visibility and control without touching operational loan data. When BTC drops 40% at 3AM, the three-tier escalation layer pauses liquidation automatically, fires Twilio SMS + email alerts to your risk officer, and blocks execution until a loan officer explicitly authorizes — every action timestamped for audit.



REGULATORY

NCUA Examiner View

Read-only audit dashboard scoped per credit union. Live on-chain collateral positions, LTV ratios, liquidation events, and full BSA/AML audit trail — built for NCUA examination from day one. Zero custody access, zero execution permissions.



Member Management

Unified member profiles combining traditional account data with on-chain wallet history, DACS eligibility scores, and loan activity — giving your team a



STABLECOIN ALLIANCE

Stablecoin Settlement Infrastructure

Three active integrations power compliant, instant settlement for



PATENT PENDING

Key Governance & Freeze Authority

Board-level cold storage with a two-tier on-chain signing

complete picture of every borrower.

credit-union loan proceeds — USDC collateral flows, cross-chain transfer protocol for multi-chain delivery, and programmable wallet infrastructure for member-owned, non-custodial wallet provisioning.

- USDC Collateral Flows
- Cross-Chain Transfer Protocol
- Programmable Wallet Infrastructure

hierarchy. Freeze authority is governed by role — loan officers, compliance teams, and board members each hold defined permissions. Compromised keys revoked instantly with no contract redeployment.

DACS — Risk Assessment Model

Twelve pillars of crypto risk assessment, backed by 10 patent-pending innovations — quantifying borrower position beyond traditional credit metrics in language the member can actually understand.

1

Wallet Age

Longevity of on-chain presence signals commitment and experience.

2

Diversification

Portfolio spread across assets reduces single-point-of-failure risk.

3

On-Chain Activity

Transaction frequency and patterns demonstrate active asset management.

4

Volatility

Historical price stability of held assets informs collateral resilience.

5

Tax-Position Score (TPSOE)

Patent-pending pillar. Scores the borrower's embedded tax liability and collateral tax efficiency — ranking pledge assets to minimize tax cost at liquidation.

6

TAM Bonus

Traditional asset weighting rewards borrowers with verified brokerage holdings via our financial data connectivity layer, enhancing the overall risk profile.

7

On-Chain Member Identity (CUMember)

PATENT
PENDING

8

Privacy-Preserving Computation (FHE-DACS)

PATENT
PENDING

9

Dynamic LTV Enforcement (DACS-LTV)

PATENT
PENDING

10

Distributed Ledger Collateral Management & Liquidation

11

XLS-66 LoanBroker Integration

PATENT
PENDING

12

Key Governance & Audit Trail

PATENT
PENDING

Patent-pending pillar. ERC-3643 compliant identity attestation — KYC verified, jurisdiction gated, and collateral-scored on-chain via ONCHAINID.	Patent-pending pillar. Member financial data is encrypted before scoring — risk assessment computations are performed without exposing raw inputs at any step.	Patent-pending pillar. DACS risk tier automatically sets each member's on-chain LTV ceiling — Tier 1 up to 70%, Tier 2 up to 65%, Tier 3 up to 55%. Enforced at the blockchain layer via ERC-3643. No manual override possible.	PATENT PENDING Patent-pending pillar. A distributed ledger architecture for real-time collateral tracking, automated margin enforcement, and liquidation control — ensuring every collateral event is recorded on-chain with full auditability and no single point of failure.	Patent-pending pillar. Native on-ledger lending via the XRP Ledger XLS-66 protocol — DACS-gated origination, dual-party authorization, and First-Loss Capital management purpose-built for NCUA-regulated credit unions.	Patent-pending pillar. Two-tier on-chain key hierarchy with instant operational key revocation and an immutable audit trail independently verifiable by NCUA examiners — no platform API access required.
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STABLECOIN INFRASTRUCTURE

Alliance Partner

Three Active Integrations

Instant, compliant settlement infrastructure — from loan origination through cross-chain delivery to the member's wallet.



USDC Collateral Flows

Loan proceeds disbursed as dollar-backed digital currency. T+0 on-chain settlement. 1:1 USD peg. Monthly attestations. Eliminates ACH delays for member disbursements.

Settlement · T+0 on-chain · Peg · USD 1:1 · Audit · Monthly



Cross-Chain Transfer Protocol

Native burn-and-mint bridge moves digital dollars across blockchains without wrapping. Zero slippage. Member receives native digital dollars on their preferred chain — Ethereum, Base, Arbitrum, Avalanche, or Solana.

Finality · ~20s cross-chain · Slippage · Zero · Chains · 5



Programmable Wallet Infrastructure

Embedded MPC-secured wallets provisioned automatically at member onboarding — no seed phrase required. Collateral held in the member's own address via smart-contract lock, satisfying NCUA beneficial-ownership standards. No omnibus account risk.

Model · Non-custodial · Keys · MPC · Compliance · NCUA beneficial-ownership

Arc

Mainnet

Q3 2026

Arc Mainnet — Q3 2026 Settlement Layer Milestone

Roadmap

Arc is the high-throughput EVM-compatible settlement layer built for institutional stablecoin flows. Aetherum will migrate USDC disbursements to Arc mainnet in Q3 2026 — delivering sub-second finality and near-zero gas costs for every credit-union loan settlement.

How It Works

1. Member deposits crypto collateral into an institutional custody wallet — held directly by the credit union, never by Aetherum.
2. DACS engine scores the borrower and sets LTV, rate, and margin thresholds.
3. Credit union issues the loan using existing infrastructure — Aetherum handles the rest.
4. Automated monitoring manages margin calls, top-ups, and liquidation if needed.

GENIUS Act Ready

The NCUA's first GENIUS Act proposed rule closed its comment period on April 13, 2026. The final implementing regulations are due by July 18, 2026 — establishing the compliance framework for how credit unions participate in the digital asset economy.

Aetherum was built inside this regulatory perimeter. Our ERC-3643 compliance modules, on-chain KYC/AML attestation, DACS risk assessment model, and institutional custody architecture are designed to operate within NCUA supervision — not around it.

Three things the GENIUS Act framework confirms about how Aetherum is built:

- **On-chain identity:** Every borrower receives an ERC-3643 compliant ONCHAINID attestation — KYC verified, jurisdiction gated, and NCUA-audit ready.
- **Custody architecture:** Each credit union holds its own institutional custody account. Aetherum is software layer only — no commingled funds, no custodial risk on the platform.
- **Compliance-first infrastructure:** DACS scoring, LTV enforcement, and collateral management logic are built to produce the audit

Revenue Model

- Platform licensing fee (annual SaaS)
- Per-loan origination fee
- DACS API access for third-party integrations
- RWA tokenization service fees (coming soon)

Credit unions keep 100% of loan interest income.

Live Integrations

- **Jack Henry FIN** — Core banking integration. 5,000+ credit unions reachable via Jack Henry ecosystem.
- **Corelation KeyStone** — Core banking integration. ~290 credit unions on KeyStone reachable from day one.
- **Fiserv AppMarket** — Core banking integration. DNA platform.
- **FIS Code Connect** — Core banking integration. IBS platform. 53M customers, 7.1M loan accounts addressable via IBS Open APIs.
- Bank-grade financial data connectivity layer
- Compliance and identity verification infrastructure
- **XRPL / XLS-66** — Native XRPL lending protocol integration. RLUSD as loan currency.
- **Pharos Atlantic Testnet** — Live testnet integration. USDC contract wired, wallet

trails NCUA examiners will expect.

- **Three-tier liquidation escalation:** Positions over \$250K or with price drops exceeding 40% are paused for mandatory human authorization — loan officers receive Twilio SMS + email alerts and action decisions from the dashboard, with every override logged for examination.

generation, balance checker, and USDC disbursement via cross-chain transfer protocol.

- **Stablecoin Infrastructure** — USDC Collateral Flows, Cross-Chain Transfer Protocol, and Programmable Wallet Infrastructure. [View details →](#)

Get In Touch

Ready to bring crypto-backed lending to your credit union?

[Schedule a 30-minute demo →](#)

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DACS (Digital Asset Credit Score) is a patent-pending risk-assessment framework developed by Aetherum. Ten patents pending. It is not a consumer credit score and is not reported to credit bureaus.

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